

Week 4, Lecture 2 WOD: Answer Key

Expectation and the Sample Mean

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. The open-ended items (1, 2b, 3b's comparison, 4c) get sample answers instead. All numbers are computed in R when this file is knitted. Throughout, $\mu = 22$ cm and $\sigma^2 = 16$ cm².

Part 1: A parameter vs. a statistic

1.

You catch $n = 5$ trout, measure them, and compute their average length $\bar{x}$. Is μ a random variable? Is $\bar{x}$ a random variable? Briefly explain your reasoning.

Answer. μ is not a random variable. $\bar{x}$ is.

Why. μ is a fixed (if unknown) property of the whole population: a **parameter**. $\bar{x}$ depends on which 5 trout happen to be caught, so a new sample gives a new value: a **statistic**, with its own distribution (which Parts 3-4 work out).

Watch for. “ μ is random because we don't know it.” Unknown is not the same as random. (This distinction returns with Bayesian inference later in the semester.)

Part 2: Working with expectations

2a.

What is $\mathbb{E}[2X + 3]$? What is $\text{Var}(2X + 3)$?

Answer. $\mathbb{E}[2X + 3] = 2(22) + 3 = 47$ cm and $\text{Var}(2X + 3) = 2^2(16) = 64$ cm².

Why. $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$. For the variance, adding a constant shifts every value by the same amount, so the spread doesn't change, and the coefficient comes out squared.

Watch for.

- $\text{Var}(2X + 3) = 2(16) = 32$ (forgetting to square the 2).
- $2(16) + 3 = 35$ (adding the constant to the variance).
- The handout's answer blanks are labeled $\mathbb{E}[2X]$ and $\text{Var}(2X)$ rather than $2X + 3$. Students who follow the blank labels will write 44 for the mean. That's a correct answer to the blank's question, so accept either and point out the mismatch.

2b.

It is always true that $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$. It is also true in this case that $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$. However, this second fact is not always true. What do you think might be the special condition that makes it true?

Answer. X and Y are **independent**.

Why. If two measurements tend to move together (e.g. two trout from the same pool, both large because the pool is productive), the spread of their sum is larger than the two spreads added. Independence rules that out.

Watch for. Answers like “they have the same variance.” That isn’t the condition. (For your reference only: the exact condition is zero covariance, which independence guarantees. Not worth raising unless asked.)

Part 3: Averaging two measurements

3a.

Rewrite M as $aX_1 + bX_2$ (i.e. what are a and b ?), then use the facts from Part 2 to derive $\mathbb{E}[M]$ and $\text{Var}(M)$ in terms of μ and σ^2 . Show every step.

Answer. $a = b = \frac{1}{2}$.

$$\mathbb{E}[M] = \frac{1}{2}\mu + \frac{1}{2}\mu = \mu \quad \text{Var}(M) = \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2 = \frac{\sigma^2}{2}$$

Why. Apply $\mathbb{E}[aX] = a\mathbb{E}[X]$ and $\text{Var}(aX) = a^2\text{Var}(X)$ to each term, then add. Adding the variances uses independence (2b).

Watch for. Not squaring the $\frac{1}{2}$, which gives $\text{Var}(M) = \frac{1}{2}\sigma^2 + \frac{1}{2}\sigma^2 = \sigma^2$: no gain from averaging at all.

3b.

Using $\mu = 22$ and $\sigma^2 = 16$ (from Part 2), what is $\text{Var}(M)$ as a number? Compare it to $\text{Var}(X_1) = 16$ alone. How does the variance of an average of two measurements compare to the variance of each of its constituent parts? $\text{Var}(M) = \underline{\hspace{2cm}}$

Answer. $\text{Var}(M) = 8$, half of $\text{Var}(X_1) = 16$.

Why. Averaging two independent measurements cuts the variance in half: an average is more precise than any single measurement.

Watch for. Answering 11 (half of μ), mixing up the mean and the variance.

Part 4: Generalizing to n measurements

4a.

Following the same logic as Part 3 (now with n terms instead of 2), derive $\mathbb{E}[\bar{X}]$ and $\text{Var}(\bar{X})$ in terms of μ , σ^2 , and n .

Answer.

$$\mathbb{E}[\bar{X}] = \sum_{i=1}^n \frac{1}{n} \mu = \mu \quad \text{Var}(\bar{X}) = \sum_{i=1}^n \frac{1}{n^2} \sigma^2 = \frac{\sigma^2}{n}$$

Why. Each term has coefficient $\frac{1}{n}$. The mean adds up n copies of μ/n . The variance adds up n copies of σ^2/n^2 (using independence).

Watch for. $\text{Var}(\bar{X}) = \sigma^2/n^2$, from squaring the $\frac{1}{n}$ but forgetting there are n terms in the sum.

4b.

Check your formula against $n = 5$, $\sigma^2 = 16$: what number does $\text{Var}(\bar{X})$ come out to? How does it compare to Part 3b when you plug in $n = 2$? $\text{Var}(\bar{X}) = \underline{\hspace{2cm}}$

Answer. $16/5 = 3.2$. With $n = 2$, $16/2 = 8$, which matches 3b.

Why. Part 3 is the $n = 2$ case of Part 4, so the general formula has to reproduce it.

Watch for. Students who get a different number at $n = 2$ than in 3b. One of the two derivations has an error; have them find it.

4c.

As $n \rightarrow \infty$, what happens to $\text{Var}(\bar{X})$? What does that mean, in plain language, about how precisely you can pin down μ with a very large sample?

Answer. $\sigma^2/n \rightarrow 0$. With enough data, the sample mean can estimate μ as precisely as you like.

Why. Each new independent measurement shrinks the variance of the average.

Watch for. “The variance of the trout lengths goes to zero.” It’s the variance of $\bar{X}$ that shrinks. Individual trout still vary by $\sigma^2 = 16$.

4d.

If $\text{Var}(\bar{X}) = \sigma^2/n$, write down the standard deviation of $\bar{X}$. How does it scale with n ?

Answer. $\text{SD}(\bar{X}) = \sigma/\sqrt{n}$, which scales as $1/\sqrt{n}$. At $n = 5$ it’s 1.79 cm.

Why. The standard deviation is the square root of the variance: $\sqrt{\sigma^2/n} = \sigma/\sqrt{n}$. So halving the standard error takes **four** times the data, not twice.

Watch for.

- Writing $\sigma^2/\sqrt{n}$ or σ/n .
- Claiming that doubling n halves the standard error.